

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2016

BIOLOGY

HIGHER 2

9648/02
22 AUGUST 2016
2 hours

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
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4	
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7	
8	
Section B	
9 or 10	
Total	100

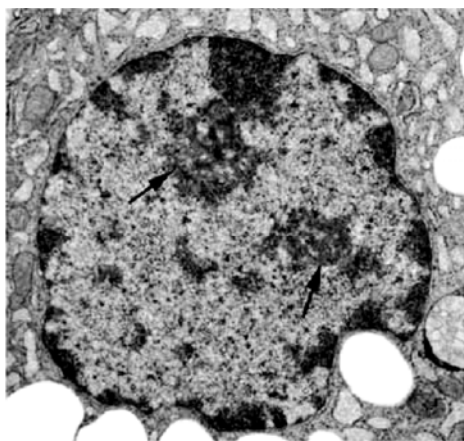
This question paper consists of **21** printed pages.

[Turn over

- 1 The appearance of cancer cells has been known to be different from normal cells. These differences have been used as a method of diagnosis by doctors. Prostate cells taken from wild-type mice and mice with prostate cancer were analysed.

Fig. 1.1 compares the same organelle found in these cells viewed under the electron microscope, while Fig. 1.2 shows the levels of ribosomal RNA (rRNA) measured in these two types of cells.

Part of a prostate cell from wild-type mice



Part of a prostate cell from mice with prostate cancer

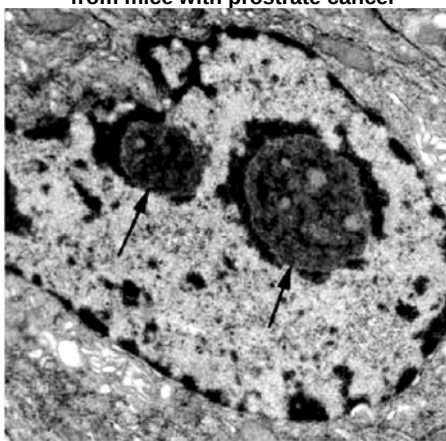


Fig. 1.1

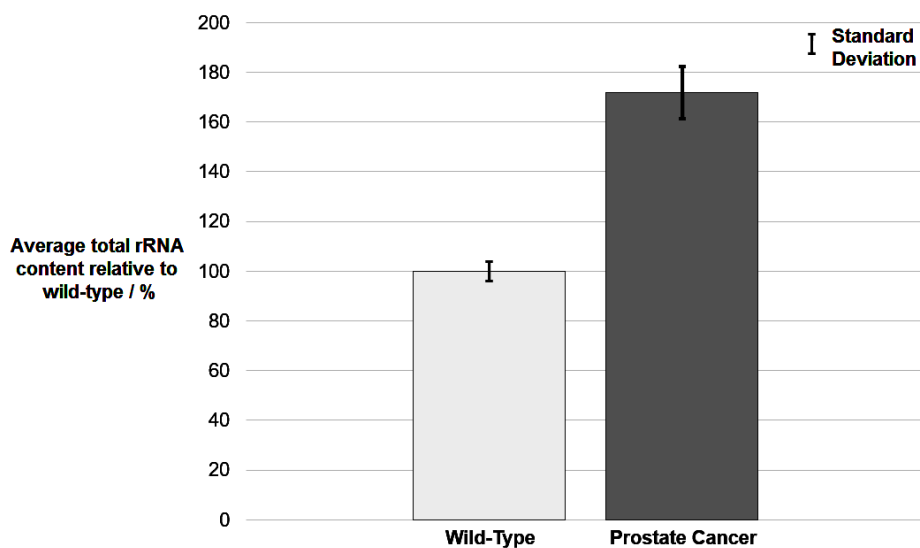


Fig. 1.2

3

(a) (i) Identify the regions indicated by the arrows in Fig. 1.1.

_____. [1]

(ii) With reference to the regions identified in Fig. 1.1 and information given in Fig. 1.2, describe the differences between wild-type prostate cells and prostate cancer cells.

_____. [2]

(iii) Explain the differences described in (a)(ii).

_____. [3]

Studies on ribosomal proteins have shown that these proteins undergo a high degree of post-translational modifications, including methylation, acetylation and phosphorylation.

(b) Suggest reasons why post-translational modification of ribosomal proteins is needed.

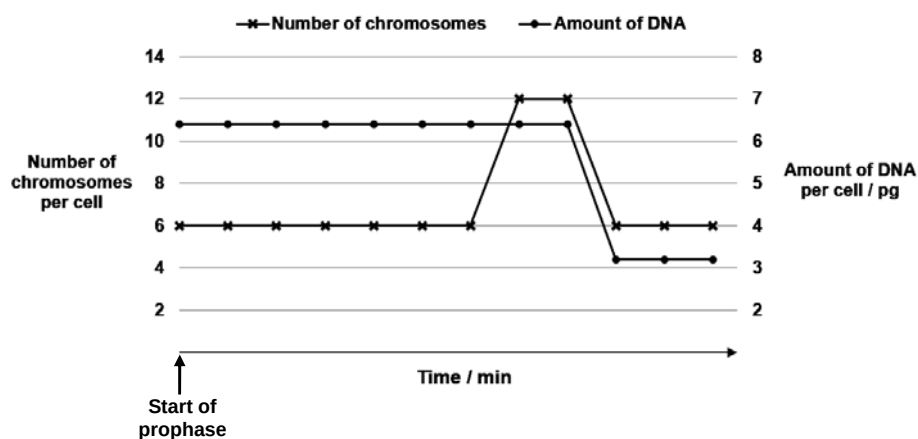
_____. [2]

[Total: 8m]

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Formatted Table

- 2 The meiotic cell cycle in the diploid germ cells of an organism can be followed by measuring the number of chromosomes as well as the amount of DNA material per cell over a period of time. Fig. 2.1 shows the results of the analyses, beginning with the start of prophase.



- (a) (i) On Fig. 2.1, indicate with an arrow where anaphase begins. [1]

- (ii) Explain whether Fig. 2.1 shows the first or the second meiotic division.

[3]

- (iii) The centromere of a chromosome comprises non-coding tandem repeats. Suggest how the structure of the centromere allows it to carry out one of its functions in cell division.

[2]

In normal cells, the ends of chromosomes do not normally fuse with each other. However, in senescent cells where the ends of chromosomes are eroded to a critical length due to the end-replication problem, chromosomes may undergo end-to-end fusions. Fig. 2.2 shows how the ends of two sister chromatids may fuse and subsequently break during cell division at a location other than the point of fusion.

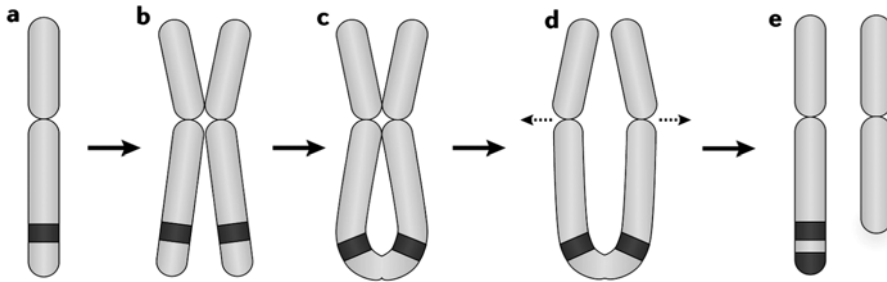


Fig. 2.2

(b) (i) Describe the result of the chromosomal mutation on the chromosomes shown in Fig. 2.2(e).

_____. [1]

(ii) Suggest how the ends of chromosomes are normally protected from end-to-end fusions.

_____. [1]

(iii) Explain how the end-replication problem will lead to the erosion of the ends of chromosomes.

_____. [3]

[Total: 11m]

- 3 (a) A new type of influenza drug has been shown to be effective against drug-resistant strains of the flu virus, according to a study led by University of British Columbia researchers in 2013. This drug works by inhibiting newly-formed influenza viruses from leaving the cell surface membrane of host cells, thus preventing the infection of other host cells.

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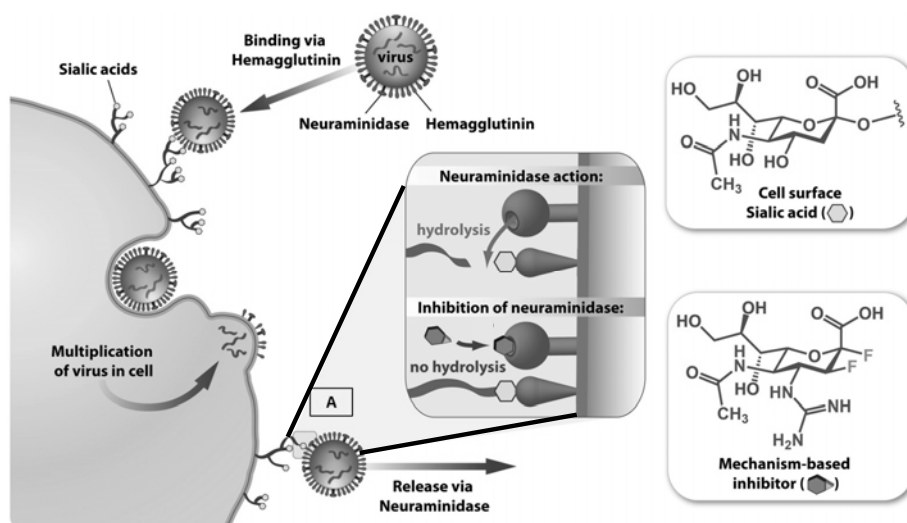


Fig. 3.1

- (i) Describe the process by which influenza viruses replicate in host cells.

[4]

- (ii) With reference to Fig. 3.1, describe how the drug prevents the newly formed influenza viruses from infecting other host cells in the process labelled **A**.

[3]

- (iii) Suggest a limitation of using this type of influenza drug.

[1]

- (b) Describe the differences between the lysogenic life cycle of a lambda phage and the life cycle of an influenza virus.

[2]

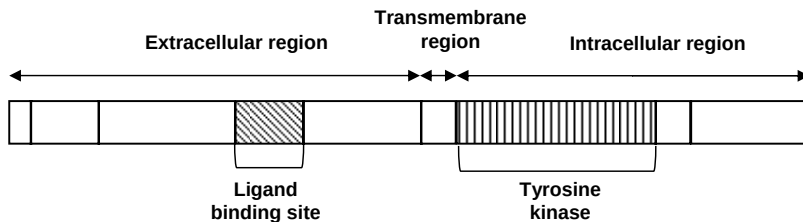
[Total: 10m]

- 4 The most common type of lung cancer is non-small cell lung cancer (NSCLC) which affects between 75% to 85% of all lung cancer patients. The *c-Met* gene which is located on chromosome 7 has been implicated in NSCLC.

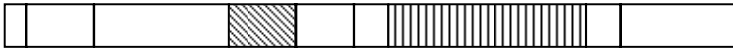
To investigate the role of *c-Met* gene in NSCLC, researchers studied the level of expression of c-Met receptor tyrosine kinase protein in several NSCLC cancer cell lines.

cDNA was generated from mRNA samples from the cancer cell lines and subsequently sequenced. Fig. 4.1a shows the normal cDNA. The corresponding protein regions it codes for are labelled. Fig. 4.1b shows two mutant cDNAs and the corresponding protein regions they code for. The corresponding regions shown in the mutant cDNA may or may not be translated to functional protein structures.

Normal cDNA:



Mutant 1 cDNA:



Mutant 2 cDNA:



With reference to Fig. 4.1a and 4.1b,

- (a) (i) State and explain the types of mutation in the *c-Met* gene that could have resulted in mutant 1 and mutant 2.

[4]

Fig. 4.2 shows the cell signaling pathway of the c-Met receptor tyrosine kinase protein.

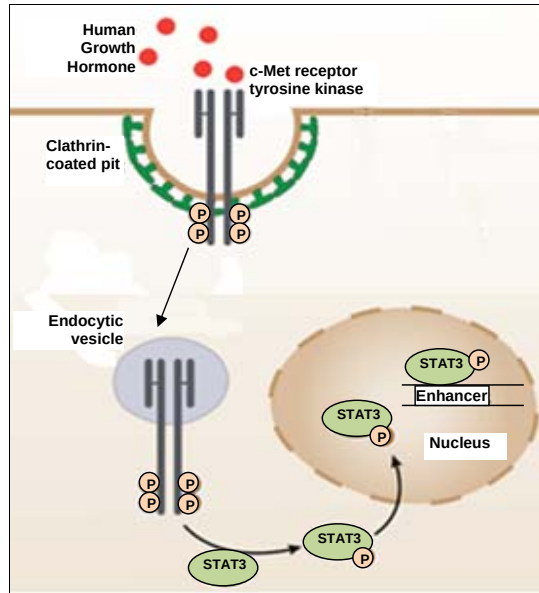


Fig. 4.2

- (b) With reference to Fig. 4.2, describe the mechanism by which STAT3 controls the expression of genes involved in cell division.

[4]

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- (c) Many cells are shown to have a mutation in the *c-Met* gene resulting in a hyperactive c-Met receptor tyrosine kinase. However, not all of these cells develop into cancerous cells. Explain why.

[4]

[Total: 12m]

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- 5 (a) A geneticist is interested in studying eye and body pigmentation in *Drosophila*. The genes for eye colour and body colour are located 11 cM apart on chromosome 3. Purple eye is produced by a dominant mutation that is lethal in a homozygous state.

Female flies with purple eyes (**E**) and ebony bodies (**b**) were mated with male flies which were true-breeding for red eyes (**e**) and yellow bodies (**B**). All the F_1 offspring of this cross had yellow bodies. F_1 female flies which had purple eyes were chosen and test-crossed with male flies. 200 flies were present in the F_2 generation. There were equal numbers between the two parental phenotypes and equal numbers between the recombinant phenotypes.

- (i) Draw a genetic diagram of the F_1 cross to show the observed results, indicating clearly the number of individuals in each different phenotypic class in the F_2 offspring.

[5]

In many other crosses, the proportion of recombinant offspring obtained is lower than the one observed in (a)(i). One possible cause is the occasional occurrence of a double cross-over in female *Drosophila*, as shown in Fig. 5.1 for a doubly heterozygous female.

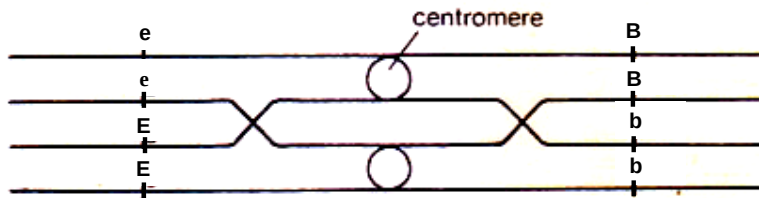


Fig. 5.1

- (ii) Assuming that a double cross-over does occur between the two genes, suggest why the recombination frequency between the two loci would be lower than the predicted percentage.

_____ [1]

In a separate experiment, the geneticist decides to investigate the chromosomal positions of genes controlling body pigmentation and wing shape in *Drosophila*. The alleles for these traits are shown below.

B: Allele for grey body **b:** Allele for black body
C: Allele for straight wings **c:** Allele for curved wings

He carried out a cross between a female fly heterozygous for both loci with a homozygous recessive black-bodied, male fly with curved wings and obtained a large number of offspring. Table 5.1 shows the results.

Table 5.1

Phenotype	Number of offspring
Grey-bodied, straight wings	35
Black-bodied, straight wings	37
Grey-bodied, curved wings	35
Black-bodied, curved wings	33

A chi-squared test was used to find out if there is any significant difference between the observed and expected ratios.

The table of probabilities is given as follows:

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

13

- (b) The calculated χ^2 value is found to be 0.229. Using the chi-squared (χ^2) test and the information given above, state the conclusion that may be drawn from the result.

[4]

[Total: 10m]

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6 Fig. 6.1 shows an electron micrograph of a plant cell.

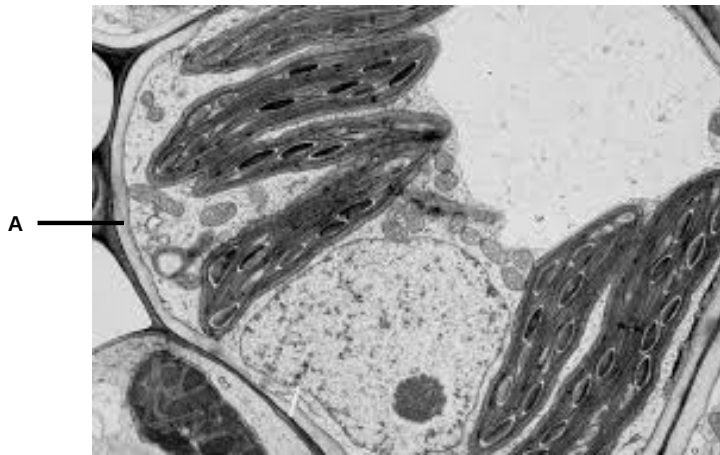


Fig. 6.1

(a) (i) Name the macromolecule which makes up structure A.

[1]

(ii) Describe how the structure of the named macromolecule in (a)(i) gives rise to its fibrous nature.

[3]

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Fig. 6.2 shows how a seed undergoes germination to form a seedling.



Fig. 6.2

(b) (i) State what is meant by respiratory quotient (RQ).

[1]

(ii) In some pea plants, the RQ has been found to be between three and four at the start of germination. Suggest a reason to explain the high RQ obtained at the early stages of germination.

[2]

(iii) Following root growth, leaves will develop and this is necessary for the seedling to harvest energy from the sun. Light intensity plays a role in determining the rate of photosynthesis. State why the rate of photosynthesis remains constant even at high light intensity.

[1]

[Total: 8m]

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- 7 The Fluid Mosaic Membrane Model of biological membrane structures was proposed as a basic framework model to describe the basic structures of biological membranes.

Synapses are unique cell junctions found between a neurone and another cell that serve to conduct signals in a neural pathway. A chemical synapse is cholinergic if it uses acetylcholine as its neurotransmitter.

- (a) Explain the significance of the Fluid Mosaic Model of membrane structures in the transmission of signals across a cholinergic synapse.

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Transmission at the neuromuscular junctions takes place in the same way as synapses between nerve cells. Fig. 7.1 shows a chemical synapse formed by the contact between a motor neurone and a muscle cell at the neuromuscular junction.

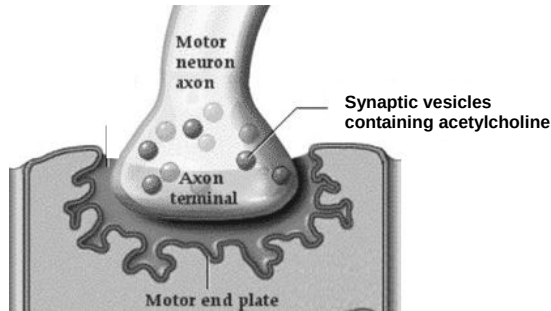


Fig. 7.1

Suxamethonium is a drug that mimics the action of acetylcholine (ACh) at acetylcholine receptors of the motor end plate. Unlike ACh, it cannot be hydrolysed by acetylcholinesterase.

- (b) With reference to the above information, explain the effect of suxamethonium on the muscles.

[4]

[Total: 8m]

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- 8 (a) The camelids are a group of mammals that share a large numbers of homologies. Recognisable camelids have been found fossilized in rock that are 45 million years old. Modern camelids have a discontinuous distribution. Guanacos and vicunas are found in South America together with the domesticated llamas and Alpacas. Bactrian Camels and Dromedaries are found in the Asian and South African deserts.

- (i) Explain how homologies in camelids supports Darwin's theory of descent with modification.

[4]

- (ii) Without considering the theory of continental drift, explain how discontinuous distribution of modern day camelids poses a problem to Darwin's theory of descent with modification.

[2]

- (b) Fig. 8.1 is a graph which shows the relationship between the number of amino acid changes of the protein haemoglobin with respect to the number of years since two organisms diverged from their common ancestor.

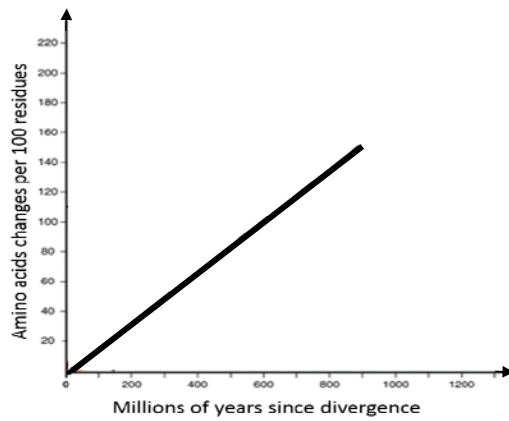


Fig. 8.1

- (i) Explain the significance of the shape of the graph shown in Fig. 8.1.

[4]

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(ii) Compare neutral mutation and silent mutation.

[3]

[Total: 13m]

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Section BAnswer **EITHER 9 OR 10**.

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.For
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use**EITHER**

- 9 (a) Using the induced-fit hypothesis, explain the mode of action of enzymes. [6]
- (b) With reference to haemoglobin, explain the significance of bonds in maintaining the protein's structure and function. [8]
- (c) Compare competitive and non-competitive inhibition of enzyme action. [6]

[Total: 20m]

OR

- 10 (a) Explain the roles of the key components of the homeostatic control system. [6]
- (b) With the use of named examples, describe the role of hormones in the regulation of blood glucose concentration in humans. [8]
- (c) Explain the significance of cell signalling pathways in homeostasis. [6]

[Total: 20m]